

Post-processing procedure for functions with logarithmic singularities

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We consider a function on $[0,1]$ with a logarithmic singularity near 0. For example,

$$u(x) = x \ln(x)$$

has such singularity at $x = 0$.

For this kind of functions, one can not directly implement the same post-processing method as in our papers. Here, we provide a way to try to reconstruct the function from Fourier information.

The main idea is the same. By introducing a map $(x \in [0,1] \rightarrow y \in [-1,1])$, we treat $u(x)$ as a function of y , i.e. $f(y) = u \circ x(y)$, which is a C^∞ function w.r.t. variable y .

$$y = -\frac{2 \ln(2)}{\ln(x/2)} - 1$$

Then we reconstruct the function using Gegenbauer polynomials $C_l^\lambda(y)$.

For a L_1 function $f(y)$, the usual Gegenbauer expansion is

$$f(y) = \sum_{m=0}^{\infty} \hat{f}^\lambda(l) C_l^\lambda(y),$$

where the Gegenbauer coefficients $\hat{f}^\lambda(l)$ are defined by

$$\hat{f}^\lambda(l) = \frac{1}{h_l^\lambda} \int_{-1}^1 (1-y^2)^{\lambda-\frac{1}{2}} f(y) C_l^\lambda(y) dy.$$

With given Fourier coefficients $\tilde{u}_k$ ($-N \leq k \leq N$) of function $u(x)$, one can obtain the truncated Fourier partial sum $u_N(x)$.

$$u_N(x) = \sum_{k=-N}^N \tilde{u}_k e^{i2\pi kx}.$$

Then we can build approximations for Gegenbauer coefficients $\hat{f}^\lambda(l)$, denoted by $\hat{g}^\lambda(l)$, as in the following.

$$\hat{g}^\lambda(l) = \frac{1}{h_l^\lambda} \int_{-1}^1 (1-y^2)^{\lambda-\frac{1}{2}} u_N(x(y)) C_l^\lambda(y) dy.$$

With m terms of Gegenbauer polynomials, we can come up an reconstructed approximation $u_N^{\lambda,m}(x)$ of $u(x)$.

$$u_N^{\lambda,m}(x) = \sum_{l=0}^m \hat{g}^\lambda(l) C_l^\lambda(y(x))$$

Since $u \circ x(y)$ is C^∞ , $u_N^{\lambda,m}(x)$ should be a good approximation with good accuracy, if the parameters λ and m are chosen properly.